



n8n and Zapier

Automate Workflows with
Zapier and n8n

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About the Speaker

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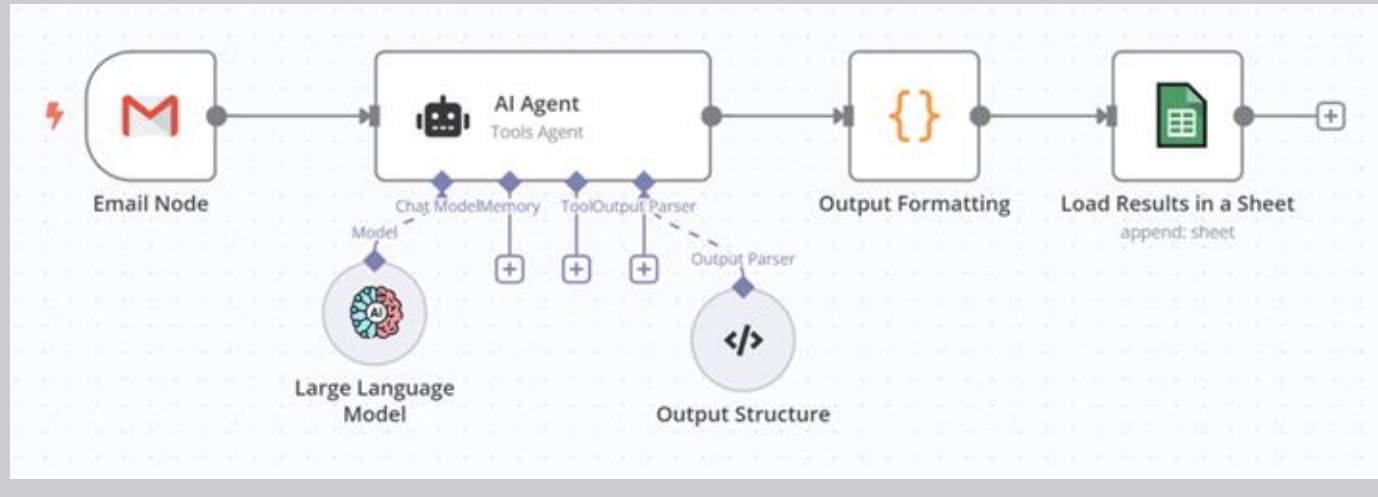
Qualification: PhD Computer Science

Experience: Several areas including but not limited to Docker, Machine Learning and Data Science, Python and much more.



Flowgramming

- Flowgramming is a specific approach to software development and education that focuses on visual logic rather than writing text-based code.



Core Concepts

- Automation is the process of connecting apps and services so they talk to each other without manual intervention. Every automation follows the same fundamental logic:
- TRIGGER → CONDITION (optional) → ACTION(s) → OUTPUT

The 4 Pillars of Workflow Logic

Triggers: The event that starts the workflow. Nothing runs without a trigger.

- Examples: "New email received", "Form submitted", "New row in spreadsheet"

Conditions / Filters: Logic gates that decide if the workflow should continue.

- Examples: "Only if email is from @company.com", "Only if amount > £500"

Actions: What happens as a result. Can be one or many steps.

- Examples: "Send Slack message", "Create CRM record", "Send email"

Data Mapping: Passing data from one step to another.

- Example: Taking the customer name from a form and inserting it into an email body.

Workflow Design Principles

Principle	Description
Idempotency	Running the same workflow twice shouldn't cause duplicate actions.
Error Handling	Always plan for failures — what happens if an API is down?
Least Privilege	Only grant the permissions your workflow actually needs.
Modularity	Break complex workflows into smaller, reusable sub-workflows.
Logging	Track what ran, when, and with what data.

Linear vs. Branching Workflows

LINEAR: [Trigger] → [Action 1] → [Action 2] → [Action 3]

BRANCHING: [Trigger] → [Condition]

└— YES → [Action A] → [Action B]

└— NO → [Action C]

PARALLEL: [Trigger] → [Split]

└— [Action A]

└— [Action B]

└— [Action C]

Triggers Comparison

Feature	Zapier	n8n
Polling interval	1–15 min (plan-dependent)	Configurable (any interval)
Webhook support	Yes (Instant triggers)	Yes (native Webhook node)
Scheduled triggers	Yes (Schedule trigger)	Yes (Cron/Schedule node)
Custom events	Limited	Full custom via code
App library	6,000+ apps	400+ native + custom HTTP

Please bear in mind that these platforms are updated constantly, so things can change.

Actions Comparison

Feature	Zapier	n8n
Multi-step actions	Yes (paid plans)	Yes (unlimited, free)
Conditional logic	Filter + Paths	IF node, Switch node
Loops/iteration	Limited (Looping by Zapier)	Full loop nodes
Code execution	Code by Zapier (JS/Python)	Code node (JS/Python)
Sub-workflows	Yes (Zap within Zap)	Yes (Execute Workflow node)
Error handling	Basic retry	Full try/catch nodes

Please bear in mind that these platforms are updated constantly, so things can change.

Decision Guide: Zapier vs n8n (My Opinion)

Use Case	Recommended
Non-technical user, quick setup	✓ Zapier
Complex logic, branching	✓ n8n
Budget-conscious / self-hosted	✓ n8n
1000's app integrations needed	✓ Zapier
AI/LLM nodes built-in	✓ n8n
Enterprise compliance/on-prem	✓ n8n
Fastest time-to-automation	✓ Zapier

Data Mapping Comparison

Zapier — Field Mapping:

- Uses a point-and-click UI with dropdown selectors
- Variables look like:

{{1. Full Name}} ← Step 1, field "Full Name"

{{2. Email Address}} ← Step 2, field "Email Address"

{{zap_meta_human_now}} ← Built-in variable for current time

Data Mapping Comparison

n8n — Expression Editor:

- Uses JavaScript expressions in double curly braces

```
{{ $json["fieldName"] }} // Current node data
```

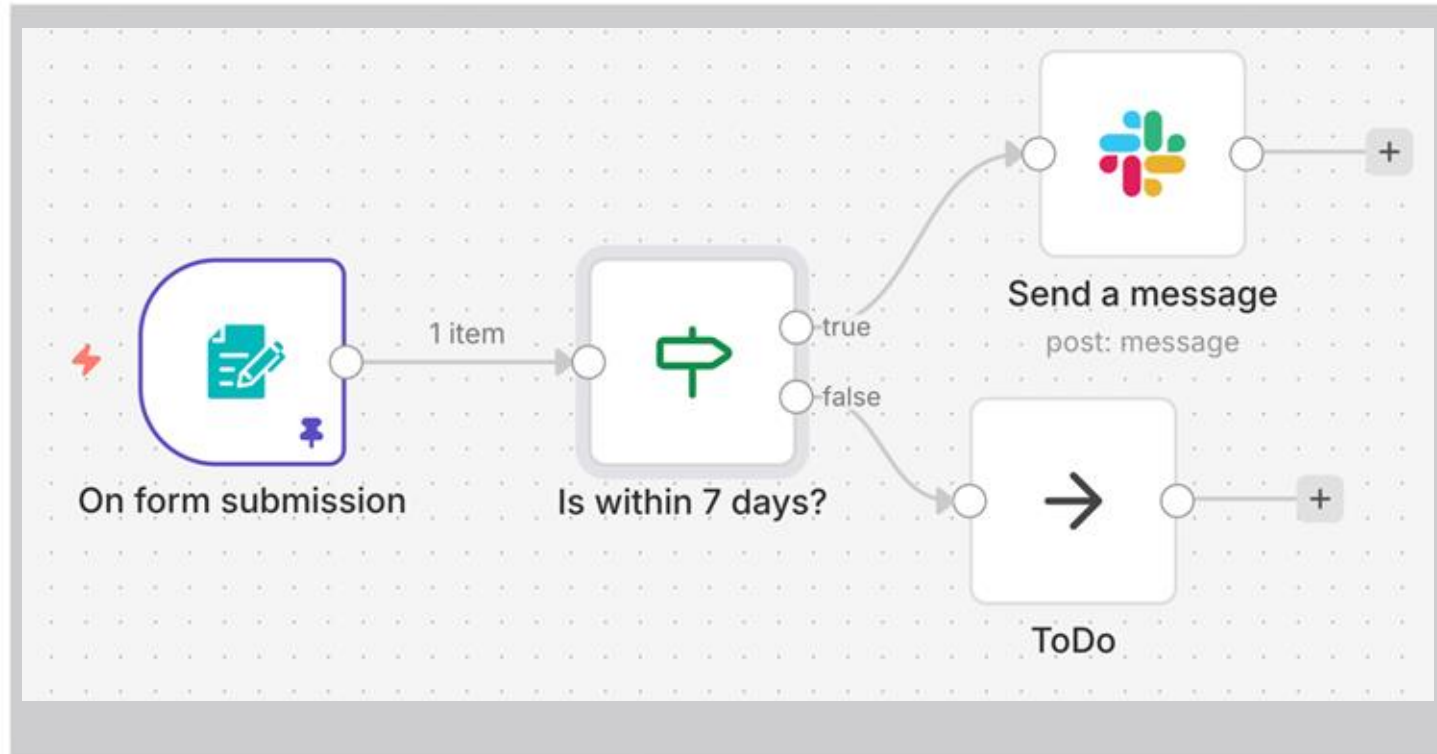
```
{{ $node["NodeName"].json["field"] }} // Specific node data
```

```
{{ $now.toFormat("yyyy-MM-dd") }} // Date formatting
```

```
{{ $items()[0].json["email"] }} // First item in array
```

Let's create a workflow on n8n

Slack Workflow on n8n



Slack Workflow on n8n

1. Set Up the Trigger (Form Submission)

- Add the Trigger: Start a new workflow and add the "On form submission" trigger node.
- Configure the Form: Give the form a title (e.g. "request a meeting") and a description.
- Add Form Fields: Create two "form elements":
 - An Email field (set type to "email" and mark as required).
 - A Preferred date field (set type to "date").
- Test the Trigger: Click "Execute Step" and fill out the test form to generate initial data.
- Pin Data: Once you have test data, click the "Pin data" button so you can continue building without refilling the form every time.

Slack Workflow on n8n

2. Add Conditional Routing Logic

- Add an If Node: Click the plus button after the trigger and select the If node from the flow section.
- Rename the Node: For clarity, rename it to something like “is within 7 days”.
- Set the Condition:
 - Value 1: Drag the “preferred install date” from the form trigger into this field to create an expression.
 - Operation: Select “is before or equal to”.
 - Value 2: Use an expression with the helper function `{{ $now.plus(7, 'days').toDate() }}` to calculate a date seven days from the current time.
- False Branch: You can add a “No Operation” (No-op) node to the false branch as a placeholder if no action is needed when the date is further than 7 days away.

Slack Workflow on n8n

3. Configure the Slack Action

- **Add the Slack Node:** From the True branch of the If node, add a Slack node and select the “Send a message” action.
- **Connect Credentials:** Select or create a Slack credential (I recommend using the OAuth route for ease of use).
- **Select Destination:** Choose a channel (e.g., “noureddin-channel”) from the list.
- **Format the Message:** Set the message text to an expression. You can combine static text with dynamic data (like the requester’s email and date) by dragging those values from previous nodes into the message box.

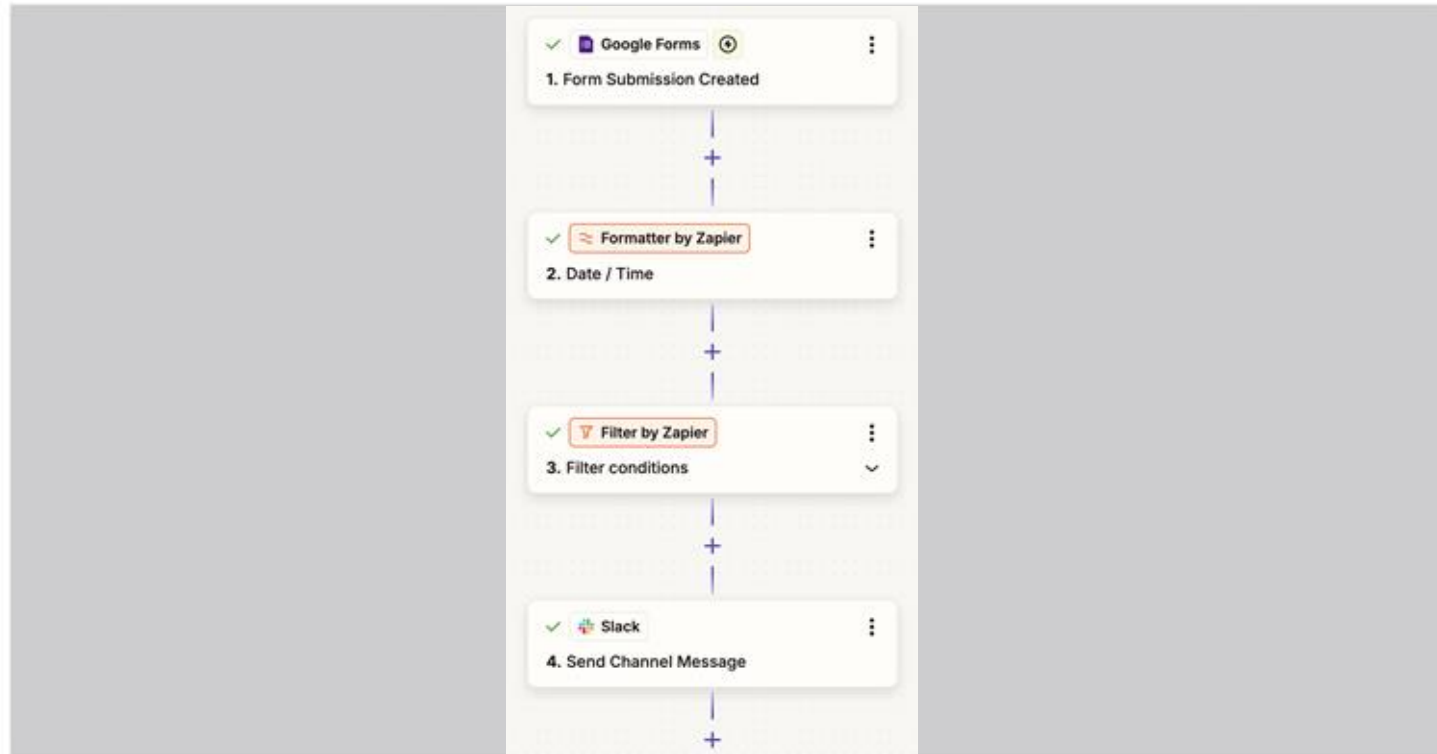
Slack Workflow on n8n

4. Test and Publish

- Run a Test: Click “Execute Workflow” to ensure the message arrives in Slack as expected.
- Save and Publish: Save your work and click Publish. The workflow is now live and will run automatically whenever someone submits the form.

Let's create a similar workflow
on Zapier

Slack Workflow on Zapier



Slack Workflow on Zapier

1. The Trigger (Form Submission)

- In Zapier: You would can a trigger from a form app like Zapier Interfaces, Typeform, or Google Forms (let's use a Google Form).
 - Ensure your form includes the same two fields: Email (set to email type) and Preferred install date (set to date type).

2. Date Calculation (Formatter)

- In Zapier: Zapier's "Filter" tool cannot perform date math (like adding 7 days) directly within the filter criteria. To replicate this, we will need to add a Formatter by Zapier step:
 - Action: Date/Time.
 - Transform: Add/Subtract Time.
 - Value: Use the Zapier "Check Time" or "Now" variable and add "+7 days".

Slack Workflow on Zapier

3. Conditional Logic (Filter)

- In Zapier: You would add a Filter by Zapier step (the equivalent of n8n's If node).
 - Condition: Set it to only continue if the "Preferred date" from your form is "(Date/Time) Before" the output date from your Formatter step.

4. The Action (Slack Message)

- In Zapier: Add a Slack action step and select "Send Channel Message".
- Configuration:
 - Connect your Slack account via OAuth.
 - Select the destination channel (e.g., "Nouredin-channel").
 - Message Text: Create an expression-style message by mapping the email and preferred install date variables from your form trigger into the message body, just as you would map dynamic data in n8n.

Slack Workflow on Zapier

5. Test and Turn On

Finally, you should test the Zap to ensure the message delivers to Slack correctly and then "Publish" it.

Example 1: Customer Booking Confirmations

Use Case: Hotel/restaurant booking system sends confirmation + adds to CRM

Workflow:

[Typeform/Calendly: New Booking]



[Formatter: Format date & name]



[Gmail: Send confirmation email to customer]



[HubSpot: Create/Update contact]



[Google Sheets: Log booking]



[Slack: Notify #bookings channel]

Example 1: Customer Booking Confirmations

Zapier Setup Highlights:

1. Trigger: Calendly → New Invitee Created
2. Action 1: Gmail → Send Email with template:
3. Subject: Booking Confirmed - {{Event Name}} on {{Start Time}}
4. Body: Hi {{Invitee Name}}, your booking is confirmed...
5. Action 2: HubSpot → Create Contact
6. Action 3: Slack → Send Message to #bookings

n8n Setup Highlights:

1. Use HTTP Request node to call Calendly webhook
2. Use Merge node to combine booking + CRM data
3. Use Send Email node with HTML template
4. Add Error Trigger node to catch failures and alert via Slack

Example 2: Helpdesk Triage

Use Case: Automatically categorize and route support tickets

Workflow:

[Email/Form: New Support Request]



[AI Node: Classify ticket] ← (n8n: OpenAI node / Zapier: AI by Zapier)



[Switch/Router: By category]

├— "Billing" → Assign to Finance team in Zendesk

├— "Technical" → Create Jira ticket + assign engineer

├— "General" → Auto-reply with FAQ link

└— "Urgent" → Page on-call via PagerDuty

Example 2: Helpdesk Triage

n8n AI Triage Setup:

// OpenAI Node - System Prompt:

"You are a support ticket classifier. Classify the following ticket into ONE category:

Billing, Technical, General, or Urgent.

Respond with ONLY the category name."

// User Message:

```
{{ $json["ticket_body"] }}
```

Example 2: Helpdesk Triage

Then use a Switch node:

Value: `{{ $json["choices"][0]["message"]["content"] }}`

Case 1: "Billing" → Zendesk node

Case 2: "Technical" → Jira node

Case 3: "Urgent" → PagerDuty node

Default: → Auto-reply node

The System Prompt (The Foundation)

The system prompt sets the governance of the AI model (i.e. agent). It is usually hidden from the end-user and acts as the “Constitution” for the model.

- Identity & Persona: “You are a senior JavaScript developer who is grumpy but helpful”.
- Capabilities & Tools: “You have access to a `search_web` function. Use it only when you lack current data”.
- Constraints: “Never output code in Python. Always keep responses under 100 words”.
- The “Vibe”: It dictates the tone, the formatting requirements, and the logic flow the agent must follow.

The User Message (The Trigger)

The user message is the input data. It is the specific problem the agent needs to solve in this exact moment.

- Task-Oriented: “How many times does ‘cat’ appear in this JSON?”
- Contextual: “Based on the last thing I said, rewrite that email to be more professional”.
- Dynamic: While the system prompt remains constant throughout a session, the user message changes with every interaction.

Quick Reference Cheat Sheet

Task	Zapier Node	n8n Node
Schedule trigger	Schedule	Cron / Schedule Trigger
HTTP API call	Webhooks / API	HTTP Request
Condition/branch	Filter + Paths	IF / Switch
Loop over items	Looping by Zapier	SplitInBatches / Loop
Run code	Code by Zapier	Code (JS/Python)
AI/LLM call	AI by Zapier	OpenAI / LangChain nodes
Format data	Formatter	Set / Function node
Sub-workflow	Zap within Zap	Execute Workflow
Error handling	Zapier Error Handler	Error Trigger node

End of Part 1

You need Credentials for AI API

Provider	Where to Get Key	Free Tier
OpenAI	platform.openai.com/api-keys	\$5 credit (new users)
Google Gemini	aistudio.google.com	Yes (rate-limited)
Anthropic Claude	console.anthropic.com	\$5 credit (new users)

Connecting AI APIs (OpenAI, Gemini, Claude) to Zapier and n8n

Connecting AI APIs in Zapier

Method 1 — Use native “AI app” integrations (fastest)

- ChatGPT (OpenAI) has a first-class Zapier app integration.
- Anthropic (Claude) also has a Zapier app integration.
- What you get:
 - Prebuilt actions (send prompt, get response, etc.)
 - Less technical setup
 - Faster to demo in training

Connecting AI APIs in Zapier

<https://zapier.com/apps/chatgpt/integrations>

<https://zapier.com/apps/anthropic-claude/integrations>

Connecting AI APIs in Zapier

Method 2 — Use Webhooks for raw API calls (most flexible)

Use this when:

- You want Gemini (or any provider) even if there isn't a perfect native Zap
- You need advanced API features (structured JSON, special parameters, custom headers)
- You want one “vendor-agnostic” design pattern

Zapier building blocks you'll typically use

- Webhooks by Zapier (POST to API)
- Formatter (clean/reshape text)
- Filter (gate logic)
- Paths (branching)
- Storage / Tables / Google Sheets / DB (persist results)
- Digest (batch daily/weekly summaries)

Connecting AI APIs in n8n

n8n supports built-in nodes for:

- OpenAI
- Anthropic (Claude)
- Google Gemini

In practice:

- Create credentials (API key) in n8n
- Use the provider node (OpenAI / Anthropic / Gemini) or chat-model sub-nodes
- Chain results into IF/Switch/DB/Slack/etc.

Keys belong in credentials vaults, not in prompts or code fields.

Customer feedback analysis pipeline with OpenAI API using Zapier

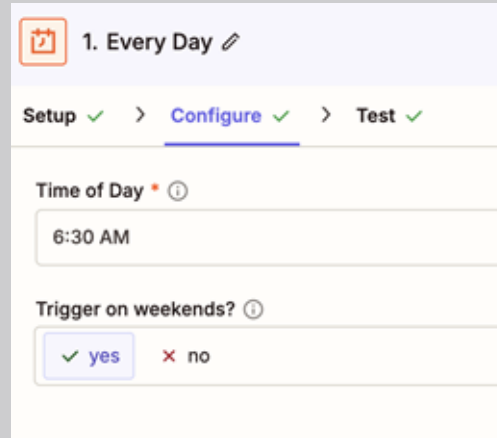
Customer feedback analysis



Customer feedback analysis

1. Set Up the Schedule Trigger

- App: Choose Schedule by Zapier.
- Event: Select "Every Day".
- Configuration: Choose a specific time.



The screenshot shows the configuration interface for a '1. Every Day' schedule trigger in Zapier. At the top, there is a header '1. Every Day' with a calendar icon and an edit icon. Below this is a progress bar with three steps: 'Setup' (checked), 'Configure' (checked and underlined), and 'Test' (checked). The main configuration area has two sections. The first section is 'Time of Day' with a red asterisk and an info icon, containing a text input field with '6:30 AM'. The second section is 'Trigger on weekends?' with an info icon, containing two radio buttons: 'yes' (selected with a green checkmark) and 'no' (unselected with a red X).

Customer feedback analysis

2. Connect Your Database (Google Sheets)

- App: Choose Google Sheets.
- Event: Select "Get many spreadsheet rows". This specific action is required because it returns lots of rows as a single JSON value, which ChatGPT can easily read.

Setup:

- Connect your Google account and select your specific Drive, Spreadsheet, and Worksheet.
 - Row Count: Increase this to ensure you capture all data.
 - First Row: Change this to 1 so that column headers are included. Headers are essential for ChatGPT to understand what the data values mean.

Customer feedback analysis

2. Get Many Spreadsheet Rows (Advanced) 

Setup  > Configure  > Test 

Drive 

My Google Drive

Spreadsheet * 

Customer-Feedback

Worksheet * 

customer-feedback

Columns * 

A:Z

Row count * 

300

Output format 

All: Rows, (JSON) Raw Rows and Formatted rows

First row 

1

Successful if no search results are found? 

Continue

Customer feedback analysis

3. Analyse Data and Draft the Report (ChatGPT)

- App: Choose ChatGPT.
- Event: Select "Conversation".
- Configuration:
 - Model: Use GPT-4 for more comprehensive analytical reports and recommendations.
 - User Message (Prompt): Define ChatGPT's role as a "senior customer review analyst". In the prompt, insert the "raw rows" dynamic variable from the Google Sheets step to provide the data for analysis.
 - Memory Key (if given the option): Assign a name (e.g., "weekly reporting") to the Memory Key field. This allows ChatGPT to remember previous chats and detect if the data is identical to the last month, preventing redundant reports.
 - Instruction Logic: You can add instructions to the prompt asking ChatGPT to generate a "no update" announcement if it receives a duplicate data set instead of drafting a new report.

Customer feedback analysis

 3. Conversation 

< Next: Step 4 >

Setup  > Configure  > Test 

Copilot suggestions

 Refresh fields

Field search

Message and Model  

Model  

gpt-4

User Message  

You are a senior data and customer review analyst. You provide insight into customer reviews.



Instructions 

Conduct an in depth analysis of the provided customer review dataset ( 2. Raw Rows: [{"Product ID", "...me by Amazon"}] ) and then generate a thorough analytical report that has your key findings and recommendations.



Customer feedback analysis

4. Distribute the Report (Slack or Teams)

- App: Choose your communication tool, such as Slack.
- Event: Select "Send Channel Message".
- Configuration:
 - Select the destination channel (e.g., "weekly reporting").
 - Message Text: Use the "Output Text" dynamic variable from the ChatGPT step. This will post the report, including titles, key findings, and recommendations, directly into the channel.

Customer feedback analysis

 4. Send Channel Message 

< Next step >

Setup  > Configure  > Test 

[Copilot suggestions](#) [Refresh fields](#) [Field search](#)

Channel 

nouredin-channel

Add Zapier app to channel automatically? 

 Yes  No



Message Text  

 3. Output Text: Based on the pr...f each product.





Send as a bot? 

 Yes  No



Bot Name 

Zapier

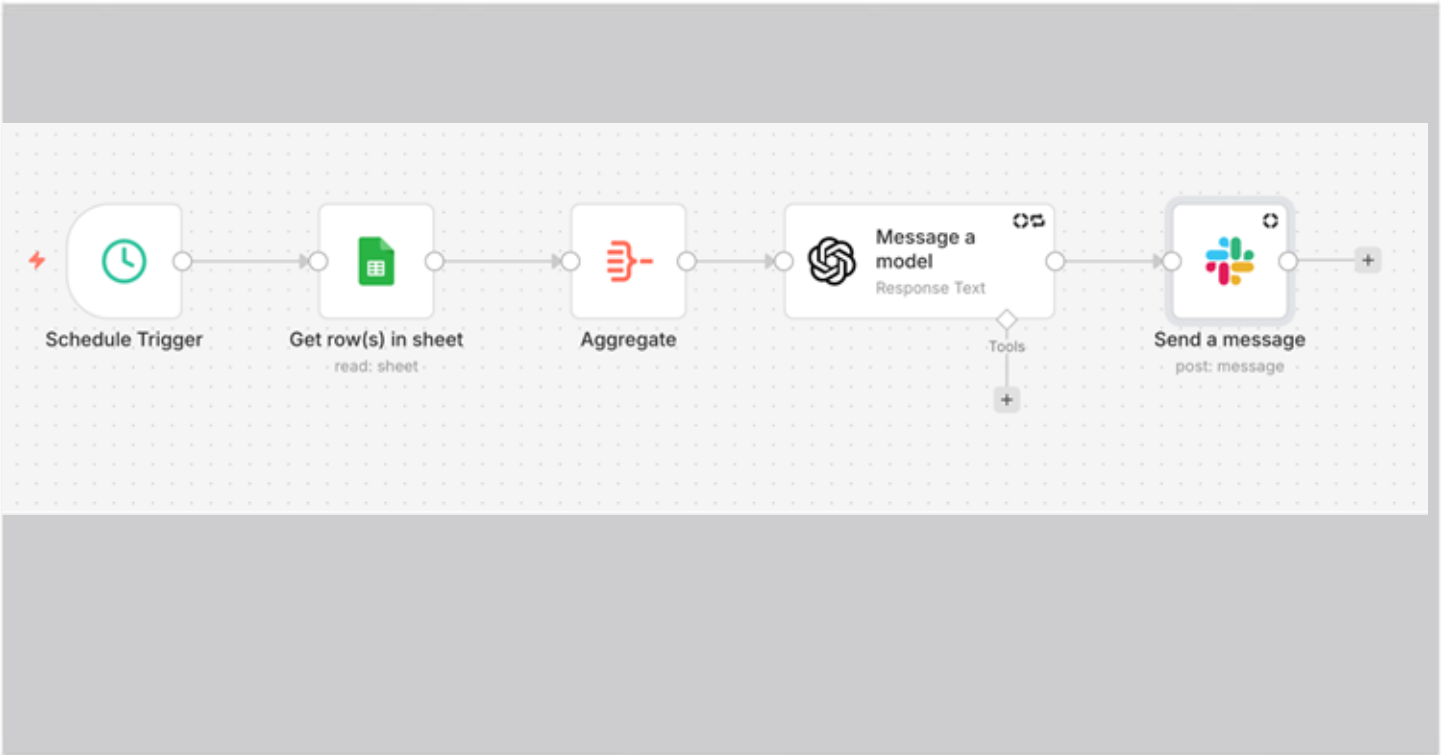


Customer feedback analysis

5. Maintenance and Activation

- Data Updates: Since the workflow uses a static spreadsheet, you must somehow update the spreadsheet (e.g. using online forms or by copying and pasting new data) before your scheduled run.
- Naming and Publishing: Name your Zap (e.g., "Customer feedback analysis from Google Sheets + ChatGPT + Slack") and click Publish to make it live.

Customer feedback analysis pipeline with OpenAI API using n8n



Customer feedback analysis

1. Add a Schedule Trigger Node

- Configure it to your preferred options.

2. Add a Google Sheets Node

- Connect to your Google Account
- Resource: Sheet within Document
- Select the Document and Sheet from List (your google sheet with the reviews should appear here)

3. Add an Aggregate Node (to aggregate column(s) into a list)

- Select Individual Fields
- In Fields to Aggregate select the column that has the reviews
- *Compare this with Zapier!*

Customer feedback analysis

4. Add an Open AI Node (Message a Model)

- Connect to your Open AI Account
- Resource: Text
- Operation: Message a Model
- Select a Model from the list
- Add this User prompt:

You are a senior data and customer review analyst. You provide insight into customer reviews.

The dataset: {{ \$json.Review }}

Conduct an in depth analysis of the provided customer review dataset and then generate a thorough analytical report that has your key findings and recommendations.

- Add this System prompt: *Only output your final report.*

Customer feedback analysis

5. Add a Slack Node

- Select Send a Message
- Connect to your Slack Account
- Resource: Message
- Operation: Send
- Send Message to: Channel
- Select your Slack Channel
- Message Type: Simple Text Message
- Message Text:

Here's a summary: {{ \$json.output[0].content[0].text }}

Support Analytics (Customer Service / Helpdesk)

What changes? Instead of reading a “Review” column, the workflow reads:

- Support tickets
- Live chat transcripts
- Email complaints
- Zendesk/Freshdesk records

What the LLM analyzes

- Sentiment (angry, neutral, satisfied)
- Issue category (billing, delivery, login problem, etc.)
- Urgency level
- Root cause patterns
- SLA (Service Level Agreement) risk signals

Example Prompt

Analyze the following customer reviews. Return JSON with:

```
{  
  "overall_summary": string,  
  "sentiment_breakdown": {"positive": number, "neutral": number, "negative":  
    number},  
  "top_themes": [{"theme": string, "count": number, "example_review_id": string}],  
  "urgent_issues": [{"review_id": string, "reason": string}],  
  "notable_quotes": [{"review_id": string, "quote": string}],  
  "recommended_actions": [string]  
}
```

Reviews: {{\$json.reviews_text}}

HR Survey Summarization (Employee Feedback / Engagement)

What changes? Instead of customer reviews, input data becomes:

- Annual engagement survey comments
- Pulse surveys
- Exit interviews
- Anonymous feedback forms

What the LLM analyzes

- Sentiment by department
- Recurring themes (workload, management, tools, morale)
- Risk signals (burnout, discrimination indicators)
- Positive culture signals

Brand Monitoring (Marketing / Reputation)

What changes? Instead of internal data, input may come from:

- Social media mentions
- Product reviews
- App store reviews
- News feeds
- Google Alerts

What the LLM analyzes

- Sentiment trends over time
- Emerging negative spikes
- Competitor comparisons
- Topic clustering (pricing, quality, customer service, ethics)

End of Part 2

HTTP Request Node in n8n

HTTP Request + Code is the “power combo” in n8n

Most “deep integrations” need at least one of these:

- A custom API endpoint (internal systems, niche SaaS)
- Non-standard auth (custom headers, HMAC, OAuth token refresh)
- Pagination and batching
- Conditional retries and error recovery
- Data transformation (reshape JSON, deduplicate, normalize)
- Multi-step workflows (create → poll → fetch → update)

<https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.httprequest/>

HTTP Request

HTTP Request gives you full API control.

Code gives you transformation + logic that sits between API calls.

Together, they turn n8n into a lightweight integration platform that can do real engineering-grade pipelines.

HTTP Request node: what it is and when to use it

Use HTTP Request when:

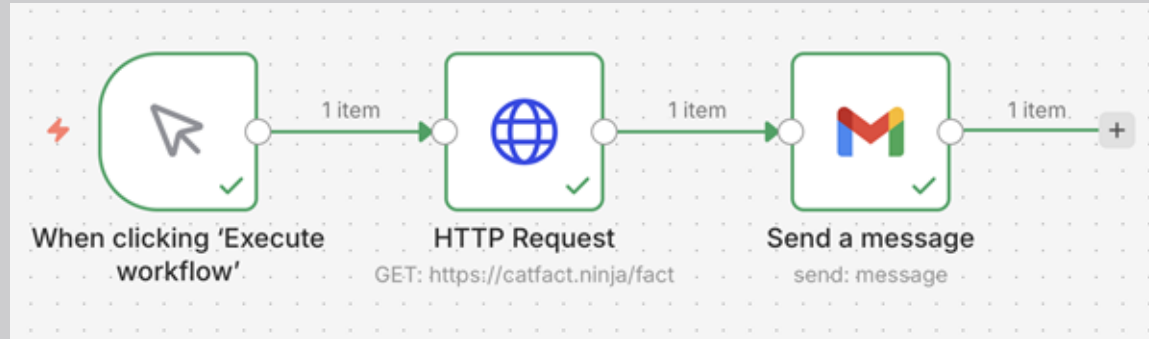
- There's no native n8n node for an app
- The native node doesn't support the endpoint you need
- You want full control over headers, query params, request bodies
- You need pagination, custom retry, rate-limit handling
- You want vendor-neutral patterns (works for almost any API)

Typical flow: HTTP Request → Code (reshape) → HTTP Request → Code (merge/enrich) → ...

Nice tutorial here: <https://www.youtube.com/watch?v=LUJzjUsJvm8>

HTTP Request node: what it is and when to use it

- Add a HTTP Request Node
 - Method: GET
 - URL: <https://catfact.ninja/fact> (this is a free API endpoint that doesn't require authentication)
- Add a GMail Node (Send a Message)
 - Configure credentials, message etc



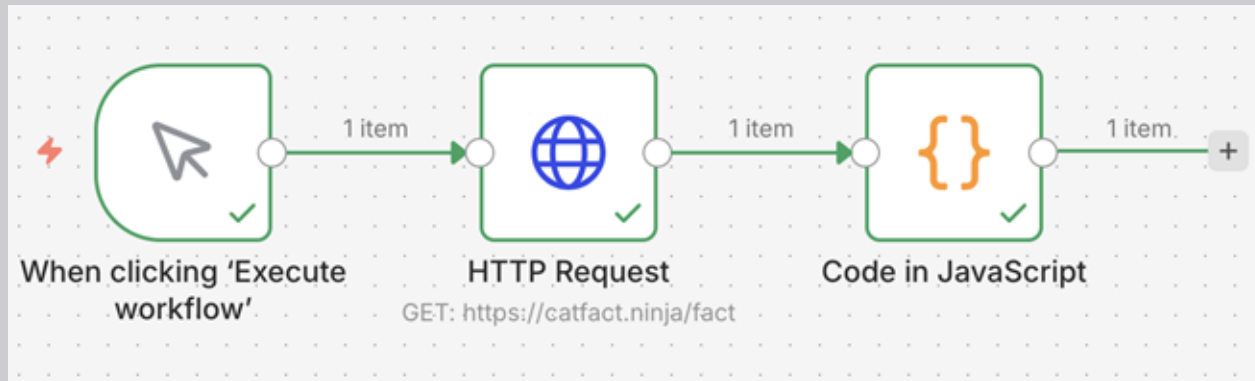
Code Node in n8n

- Use the Code node to write custom JavaScript or Python and run it as a step in your workflow.
- **Use it when you need to do something a regular node cannot do!**
- The Code node is your:
 - data shaper
 - validation layer
 - control logic engine
 - “glue” between different systems’ data formats

<https://docs.n8n.io/code/code-node/>

Code Node in n8n

- Go back to the HTTP Request Workflow, replace the GMail Node with a Code Node (Javascript).
- Use the Code on the next slide to return how many times the word “cat” appears in the text.



Code Node in n8n

```
// Access the text from the fact field
const factText = $input.first().json.fact;

// Use a regular expression to find all instances of "cat" (case-
insensitive)
// The 'g' flag finds all matches, and the 'i' flag makes it case-
insensitive
const matches = factText.match(/cat/gi);

// Get the count (handle null if no matches are found)
const count = matches ? matches.length : 0;

console.log(`The word "cat" appears ${count} times.`);

return {count};
```

The Big Picture: What Is Multi-App Automation?

Multi-app automation connects:

1. Data sources (CRM, Forms, Helpdesk, Sheets, APIs)
2. Processing layers (AI models, enrichment APIs, business logic)
3. Action layers (Slack, email, dashboards, databases, ticketing systems)

The architecture is the same across tools:

Trigger → Collect Data → Enrich → Analyze → Decide → Act → Store

Zapier and n8n both support this — but in slightly different styles.

Data Enrichment (Adding Intelligence to Raw Data)

What is Data Enrichment? Data enrichment means:

- Enhancing raw input with additional information from other systems or AI.

Example:

- A new lead submits a form.
- You enrich it with:
- Company size (via API)
- Industry classification
- Website summary (via AI)
- Sentiment from prior emails
- Risk score

Zapier

Zapier works best for:

- Linear workflows
- Standard SaaS integrations
- Business-friendly automation

Zapier strengths:

- Very fast setup
- Strong SaaS ecosystem
- Easy branching with Paths

Limitations:

- Less control over looping and complex pagination
- Less expressive data transformation

Zapier - Typical Pattern

Trigger: New form submission / new CRM record.

Step 1:

Call an enrichment API via:

- Built-in app (e.g., Clearbit-like service)
- Webhooks (custom API call)

Step 2:

Use ChatGPT integration to:

- Classify lead quality
- Summarize company website
- Extract key signals

Step 3:

- Update CRM with enriched fields

Step 4:

Route:

- High value → notify sales
- Low value → nurture campaign

n8n excels when:

- You need multiple APIs
- You need complex logic
- You want structured JSON manipulation
- You want partial engineering control

Zapier strengths:

- Deep API flexibility
- Strong branching logic
- Better for technical teams

n8n - Typical Pattern

Trigger: Webhook / CRM event /
Sheet row

Step 1:

- HTTP Request → enrichment API
- Code node → normalize output

Step 2:

- OpenAI node → classify / summarize

Step 3:

- Merge enrichment + AI analysis

Step 4:

- Switch node → route by risk score

Step 5:

- Update CRM
- Write to database

Report Generation (Turning Data Into Insight)

Multi-app automation becomes powerful when you move from reacting to events to generating insight.

Automated Operational Reports

Use Case - Daily summary of:

- Support tickets
- Customer reviews
- Sales leads
- Website feedback

Workflow Pattern

1. Trigger (daily cron)
2. Pull data (Sheets, CRM, Helpdesk API)
3. Aggregate
4. Send to LLM for structured summary
5. Format for Slack / Email
6. Store in archive (Notion/DB/Drive)

Zapier

Zapier approach:

- Trigger: Schedule by Zapier
- Pull rows from Sheets
- Use ChatGPT to summarize
- Use Digest to combine multiple records
- Send to Slack channel

Best for

- Executive summaries
- Simple digests
- Marketing and support dashboards

n8n method:

- Cron trigger
- HTTP Request (pull tickets with pagination)
- Code (aggregate + count themes)
- OpenAI (structured JSON output)
- Code (format Slack message)
- Slack node
- Save structured report in DB

Better for

- Advanced metrics
- Trend analysis
- Multi-step analysis
- Error handling

AI-Driven Decisions (Automation That Thinks)

This is where automation becomes strategic.

Instead of just:

“If X happens → notify someone”

You do:

“If X happens → analyze → score → decide → act automatically”

Example Decision Types You Can

Automate

A) Risk scoring

- Customer email tone + account size → churn risk
- Supplier message tone → contract risk
- Brand spike in negative mentions → crisis alert

B) Priority routing

- High urgency → escalate
- Medium urgency → queue
- Low urgency → archive

C) Opportunity detection

- Positive sentiment + high value customer → send upsell alert
- Social post praising product → notify marketing

Zapier supports AI-driven decisions through:

- ChatGPT integration
- Paths (branching)
- Filters
- Formatter

Example:

- New review
- ChatGPT: classify sentiment + urgency
- Path A (urgent negative) → create support ticket
- Path B (positive) → add to testimonial list
- Path C (neutral) → store only

Simple, fast, business-oriented.

n8n allows deeper AI orchestration:

- Multi-model strategy (cheap model first, expensive model only if needed)
- Confidence thresholds
- Retry logic if model uncertain
- Loop through items individually
- Combine multiple signals

Example:

- Pull 100 tickets
- Code → split high-risk keywords first
- Only send suspicious ones to LLM
- If LLM confidence < 0.6 → send for human review
- Update CRM status automatically

This becomes semi-autonomous AI workflows.

Zapier vs n8n for Multi-App Automation

Capability	Zapier	n8n
Ease of setup	★★★★★	★★★
API flexibility	★★	★★★★★
Complex branching	★★★	★★★★★
Data transformation	★★	★★★★★
Enterprise-style workflows	★★	★★★★★
Business team usability	★★★★★	★★★

Hands-On Lab: Multi-model
AI workflow — Gemini for
planning, GPT for execution,
Claude for review.

End of Part 3

Seven Important Aspects:

- 1- Deployment Model
- 2- Governance & Access Control
- 3- Monitoring & Observability
- 4- Error Handling & Recovery
- 5- Security & Data Residency
- 6- Scalability
- 7- Cost Structure

1- Deployment Model

Zapier Teams

Deployment Type: Fully managed SaaS

- No infrastructure to manage
- Hosted entirely by Zapier
- Automatic updates
- No server configuration

Best for:

- Business-led automation
- Non-technical teams
- Quick time-to-value

What this means:

- Fast to start
- Minimal IT involvement
- Limited infrastructure customization

Deployment Type: Managed cloud by n8n

- Hosted by n8n
- Infrastructure abstracted away
- Automatic upgrades

You get:

- More flexibility than Zapier
- Still low DevOps overhead
- Control over workflows but not servers

Best for:

- Technical teams
- AI-heavy workflows
- Moderate customization needs

n8n Self-Hosted

Deployment Type: You host it (Docker, Kubernetes, VM, etc.)

You Control:

- Server environment
- Networking and Security rules
- Scaling strategy, Database and Backups

You must manage:

- Updates, Monitoring, Reliability and Security patches

Best for:

- Enterprises
- Regulated industries
- Companies requiring data residency control
- Advanced internal AI orchestration

2- Governance & Access Control

Zapier Teams

Provides:

- Shared folders
- Role-based access
- Admin controls
- Approval workflows (depending on plan)
- Centralized credential management

Limitations:

- Less granular than engineering platforms
- No deep environment separation (dev/staging/prod not native)

Good for:

- Marketing, Sales, Ops teams

Provides:

- Role-based access
- Credential management
- Project separation
- Workflow sharing

Better suited for:

- Engineering-style collaboration
- Controlled access to API keys
- Structured team workflows

n8n Self-Hosted

Fully customizable:

- Integrate with SSO (Azure AD, Okta)
- Custom RBAC
- Separate dev/staging/prod environments
- Network-level restrictions
- Internal-only deployment

But requires:

- IT team involvement

Enterprise advantage:

- Complete governance control

3- Monitoring & Observability

My opinion: This is where the differences become more significant.

Zapier Teams Monitoring

You get:

- Task history
- Error logs
- Basic run logs
- Task usage metrics
- Email notifications for failures

What you don't get:

- Deep infrastructure metrics
- Advanced log streaming
- Custom monitoring pipelines

Monitoring style:

- Business-friendly
- Dashboard-level visibility

Limitations:

- Harder to implement advanced alerting logic
- Limited log export

n8n Cloud Monitoring

You get:

- Execution logs
- Per-node error visibility
- Manual re-run capabilities
- Error workflows (trigger on failure)
- Environment variable visibility

Better for:

- Technical debugging
- Multi-step workflow tracing

Limitations:

- Still limited compared to self-hosted observability tools.

n8n Self-Hosted Monitoring

This is where things become enterprise-grade.

You can integrate:

- Prometheus
- Grafana
- ELK stack
- Cloud monitoring (AWS CloudWatch, Azure Monitor, etc.)
- Centralized logging systems

You can monitor:

- CPU usage, Memory, Queue performance, Workflow execution time
- Failed runs, API latency, Retry counts

This enables:

- True production-grade monitoring
- SLA enforcement
- Performance tuning
- Cost optimization

4- Error Handling & Recovery

Zapier Teams

You get:

- Automatic retries (limited)
- Error email alerts
- Zap stops after repeated failures
- Manual fix required

Less ideal for:

- Multi-step AI pipelines with dependencies

Works well for:

- Linear workflows
- Low-complexity automation

n8n (Cloud & Self-Hosted)

You can design:

- Explicit retry branches
- Error-catching workflows
- Dead-letter queues
- Manual review loops
- AI confidence thresholds

Example:

- If LLM output invalid JSON:
 - Retry
 - If still invalid → route to Slack alert
 - Pause workflow

This is true workflow resilience.

5- Security & Data Residency

Zapier Teams

- Data flows through Zapier servers
- Good general compliance posture
- Limited control over where data resides

Concern:

- Sensitive AI data may pass through third-party infrastructure.

n8n Cloud and Self-Hosted

n8n Cloud

- Hosted by n8n
- Similar SaaS model
- Less ecosystem exposure than Zapier
- Still third-party hosted

n8n Self-Hosted

Major advantage:

- Data never leaves your infrastructure (unless calling external APIs)
- You control encryption
- You control region
- Internal-only network access possible

Ideal for: Healthcare, Finance, Government, Oil & gas, Highly regulated industries

6- Scalability

Scalability

Zapier Teams

Scales via:

- Plan upgrade
- Task limits

But:

- Pricing tied to task count
- High-volume AI workflows can become expensive

n8n Cloud

Scales via:

- Plan tier
- Execution limits

Better flexibility than Zapier for complex flows.

n8n Self-Hosted

You scale via:

- Horizontal scaling (Kubernetes)
- Worker queues
- Distributed architecture

This supports:

- Large-scale AI orchestration
- Internal automation platforms

But requires: DevOps expertise

7- Cost Structure

Cost Structure

Zapier Teams

- Task-based pricing
- Can become expensive with:
 - High-frequency triggers
 - AI calls
 - Multi-step workflows

Best for:

- Moderate usage
- Business

automation

n8n Cloud

- Subscription model
- More cost-efficient for complex workflows
- Less per-task cost pressure

n8n Self-Hosted

- Infrastructure cost
- DevOps cost
- Potentially cheapest at scale
- Most control